SPARTA GLOBAL

AI FUNDAMENTALS.

COURSE SPECIFICATION
AI & TECH INSTITUTE

A five-session programme that takes professionals from understanding how AI systems are built to delegating real work to AI agents — with the judgement, oversight and accountability that responsible use demands.

THE PROGRAMME

COURSE AT A GLANCE.

AI Fundamentals builds practical, transferable AI literacy. Learners move from foundational understanding to applied delegation across five progressive sessions, each combining taught content, live practical work and an assignment applying it.

5	13	5	100%
PROGRESSIVE SESSIONS	LEARNING OBJECTIVES	APPLIED ASSIGNMENTS	PRACTICAL COMPONENT

WHO IT IS FOR

Professionals in technical and non-technical roles who need to use AI credibly at work: to understand what these systems are doing, judge when to trust them, and apply them to real tasks under appropriate oversight. No prior AI or machine learning knowledge is assumed.

STRUCTURE

Each session opens with its position in the arc and its learning objectives, works through three or four taught topics, includes a live practical component and group discussion, and closes with an assignment applying the session's content.

ON COMPLETION, LEARNERS CAN

- Explain how AI systems and large language models are built, trained and used
- Recognise the failure modes of AI systems and apply the right mitigation to each
- Engineer prompts and context that produce reliable, verifiable outputs
- Choose the appropriate model and tool for a task, weighing accuracy, risk and cost
- Design automated and AI-powered workflows with human oversight built in
- Delegate work to AI agents and independently verify what came back

> ***"Delegate the work. Don't delegate ownership of the work."***
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PROGRESSION

THE COURSE ARC.

The five sessions are sequential and cumulative. Each one depends on the concepts established before it, and the level of autonomy learners are asked to manage increases as the course progresses.

	SESSION	FOCUS	ASSIGNMENT
01	**Demystifying AI** How it works and works for you	What AI is, how it is trained, and how a model is built and improved	Multiple-choice knowledge check
02	**Understanding Modern AI** LLMs, risks and considerations	How large language models generate responses, and where they fail	Applied parameter task or written case analysis
03	**Effective AI Use** Prompt engineering & responsible practice	Guiding AI deliberately, and using it responsibly	Applied breakout exercise and team review
04	**AI for Automation** Workflows & AI-powered processes	Turning judgement into repeatable, accountable processes	Built workflow with documented reasoning
05	**Agentic AI** From instructions to intent	Delegating goals rather than steps, and verifying the result	Final delegated project with cost and risk analysis

THREADS THAT RUN THROUGHOUT

Responsible use, human-in-the-loop oversight, output verification and the AI Fluency 4Ds — Delegation, Description, Discernment and Diligence — are introduced early and revisited in every subsequent session as the stakes rise.

KEPT CURRENT BY DESIGN

Taught concepts are stable; the tools are not. Practical components, worked examples and demonstrations are refreshed each cohort to reflect the current tooling landscape, without altering the learning objectives.

SESSION 01

DEMYSTIFYING AI.

How it works and works for you — exploring how AI systems are built and how they operate, from neural networks and machine learning models to practical applications and opportunities.

LEARNING OBJECTIVES

- Confidently define AI and its subfields
- Understand how AI is powered, and which parameters and techniques are crucial to its success

MAIN TOPICS

01 INTRO TO AI

What AI is, where it already appears in daily life and industry, and how the key terms fit together — AI, machine learning, deep learning and generative AI.

02 THE BRAIN OF AI

How data becomes decisions: neural networks and their common architectures, the difference between an architecture and a trained model, the three training methods, and inference.

03 CLASSIFICATION AI WALKTHROUGH

A model built, trained and used end to end — datasets and labelling, training runs, accuracy metrics and the retraining loop.

SESSION COMPONENTS

TAUGHT CONTENT | DISCUSSION PROMPTS | LIVE PRACTICAL WALKTHROUGH

ASSIGNMENT: KNOWLEDGE CHECK

Assignment: a ten-question multiple-choice knowledge check evaluating understanding of terminology, training approaches and model behaviour.

SESSION 02

UNDERSTANDING MODERN AI.

LLMs, risks and considerations — investigating the models that define modern AI, from the architectures that power them to the strengths, limitations and reliability challenges of AI systems.

LEARNING OBJECTIVES

- Confidently explain how LLMs work, including the architecture and concepts that underpin their capabilities
- Identify the strengths, limitations and common failure modes of AI systems, and the approaches used to improve their performance and reliability

MAIN TOPICS

01 INTRO TO LLMS

What large language models are, what they generate, and how they differ from multimodal models.

02 HOW LLMS WORK

Why an LLM predicts rather than retrieves: tokenisation, embedding, transformer layers, and how probabilities become a response.

03 PERFORMANCE AND OPTIMISATION

Tuning model behaviour through the four inference parameters, and the degree of control open and closed models expose.

04 CAPABILITY CONSIDERATIONS

Where AI systems fall short — hallucinations, bias, data quality, tokenisation and context limits, model collapse — and the mitigation for each, including RAG.

SESSION COMPONENTS

TAUGHT CONTENT | REFLECTION TASK | LIVE PRACTICAL DEMONSTRATION

ASSIGNMENT: APPLIED CHOICE

Assignment: either a documented experiment modifying inference parameters and evidencing the effect of each change, or a written analysis of a real-world AI failure covering cause, harm and future mitigation.

SESSION 03

EFFECTIVE AI USE.

Prompt engineering and responsible practice — developing the skills to guide AI effectively through prompt and context engineering, while applying critical judgement and responsible use principles.

LEARNING OBJECTIVES

- Understand how structured prompts and contextual inputs improve the quality and reliability of AI-generated outputs
- Make informed decisions about tool selection, output validation and risk management when using AI

MAIN TOPICS

01 AI BEHAVIOURAL TRAITS

Why systems optimised to be helpful can mislead: sycophancy, recency bias and overconfidence, and how to keep each in check.

02 PROMPT AND CONTEXT ENGINEERING

What separates a basic prompt from an engineered one — persona, task, context, constraints — and designing the wider information environment an AI works within.

03 RESPONSIBLE AI USE

Data protection and confidentiality, environmental impact, human-in-the-loop oversight, model selection against risk and cost, and the AI Fluency 4Ds.

SESSION COMPONENTS

TAUGHT CONTENT | MODEL SELECTION EXERCISE | TEAM BREAKOUT EXERCISE

ASSIGNMENT: TEAM BRIEFING

Assignment: a timed team exercise producing a briefing from a supplied information set, presented back and reviewed against responsible-use benchmarks — relevance, verification, proportionate context and exclusion of sensitive information.

SESSION 04

AI FOR AUTOMATION.

Workflows, automation and AI-powered processes — identifying opportunities for workflow automation, designing AI-powered processes, and implementing appropriate human oversight and safeguards.

LEARNING OBJECTIVES

- Identify opportunities to automate tasks and processes using workflows and AI
- Describe how data, automation and AI work together within modern workflow platforms
- Design responsible AI-powered workflows that balance efficiency with accountability

MAIN TOPICS

01 UNDERSTANDING WORKFLOWS

What a workflow is, and its anatomy — triggers, inputs, steps, decisions and outputs — against everyday and workplace examples.

02 AUTOMATED WORKFLOWS

What should and should not be automated, the no-code platform landscape and its trade-offs, and how connectors inherit their builder's permissions.

03 AI-POWERED WORKFLOWS

What changes when a model, rather than a rule, decides a step — and how to assign each step in a process to a rule, a model or a person.

04 RESPONSIBLE AUTOMATION

Why automation fails at scale, at speed and in silence, and how to design approval, escalation, accountability and verification into it.

SESSION COMPONENTS

TAUGHT CONTENT | GUIDED WALKTHROUGH | GROUP DESIGN ACTIVITY

ASSIGNMENT: BUILD A WORKFLOW

Assignment: build a working automated workflow on a platform of choice that solves a realistic problem, including at least one automated action, at least one AI-powered step and appropriate human oversight — submitted with documented reasoning for each step and evidence of live runs.

SESSION 05

AGENTIC AI.

From instructions to intent — understanding how agentic AI evolved to set its own steps toward a goal, and what that changes about oversight, trust and control.

LEARNING OBJECTIVES

- Describe how enterprise AI has evolved
- Understand how agents can be applied to a task
- Explain how AI agents plan and act towards their goal
- Assess the real cost of agentic delegation

MAIN TOPICS

01 THE EVOLUTION OF ENTERPRISE AI

The shift from assistance to delegation, what an AI agent actually is, and four tests for judging whether a tool is genuinely agentic.

02 HOW AI AGENTS WORK

The anatomy of an agent, the plan, act and check loop it runs, and the five things any agent needs: mission, context, tools, permissions and instructions.

03 AGENTS IN PRACTICE

Agents applied to knowledge work, to business applications and to code — why coding agents matured first, and how vibe coding leaves understanding behind.

04 RESPONSIBLE AGENCY

Blast radius, least privilege, reversibility, safe environments and independent verification — plus five questions to answer before delegating any task.

SESSION COMPONENTS

TAUGHT CONTENT | LIVE AGENT DEMONSTRATIONS | APPLIED CLASSIFICATION EXERCISE | FINAL ASSIGNMENT

Assignment: complete a realistic business or coding task by delegating it to an AI agent, submitting the resulting artefact, evidence of the agent's work including a point of human intervention, a delegation review, a justification for choosing an agent over a workflow, and a cost and risk analysis.

ASSIGNMENTS AND DELIVERY

HOW LEARNING IS EVIDENCED.

Assignments escalate with the course. Early sessions confirm understanding; later sessions require learners to produce working artefacts and defend the decisions behind them.

	ASSIGNMENT	WHAT IT EVIDENCES	TYPE
01	Knowledge check	Command of core terminology, training approaches and model behaviour	Multiple choice
02	Applied task or case analysis	Understanding of model behaviour and of how AI failures arise and are mitigated	Practical or written
03	Briefing exercise	Prompt and context engineering under time pressure, plus responsible-use judgement	Team practical
04	Workflow build	A working process combining rules, AI judgement and human oversight, with reasoning	Built artefact
05	Delegated project	Effective delegation to an agent, independent verification, and cost and risk analysis	Final project

DELIVERY NOTES

Sessions are instructor-led and delivered in sequence, each combining taught content with live practical work and structured discussion. Group activities and breakout exercises are used from Session 3 onward, where the material shifts from understanding AI to applying it.

Learners need access to an approved enterprise AI tool for the applied sessions, and to a workflow platform and an agentic tool for the final two assignments.

SCOPE OF THIS SPECIFICATION

This document specifies the structure, learning objectives, taught topics and assignments of the course. Individual demonstrations, worked examples and tool choices are deliberately out of scope: they are refreshed regularly so that delivery keeps pace with a fast-moving landscape, while the objectives and topics above remain the constant.

> ***"Use AI to move faster through work you understand — never to skip understanding it."***
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